called *diffusion processes.* Thus, the steady state case is a solution of a non-divergence form equation.

We develop the basic theory of elliptic equations in divergence form, recasting the most common boundary value problems within the functional framework of the abstract variational problems of Section 7.4. This functional setting turns out to be very well adapted to implement numerical approximation procedures, such as Galerkin method.

8.2 The Poisson Problem

Assume we are given a domain $\Omega \subset \mathbb{R}^n$ and two real functions a_0, $f : \Omega \to \mathbb{R}$. We want to determine a function u satisfying the equation

$$-\Delta u + a_0 u = f \qquad \text{in } \Omega$$

and one of the usual boundary conditions on $\partial\Omega$.

The Poisson problem models a variety of stationary phenomena. For instance, if $n = 2$, u can be interpreted as the equilibrium position of a perfectly elastic membrane under a distributed charge f.

A Dirichlet condition ($u = g$ on $\partial\Omega$) means that the boundary of the membrane keeps an assigned position. A Robin condition ($\partial_{\boldsymbol{\nu}} u + \beta u = g$ on $\partial\Omega$, $\beta > 0$) corresponds to a membrane whose boundary is subject to an elastic restoring force. When $\beta = 0$ (Neumann condition) the membrane boundary is free to move along a vertical guide.

When u describes the equilibrium concentration of a substance, a Dirichlet condition prescribes the boundary concentration, while a Neumann condition prescribes its flux through the boundary.

Let us examine what we mean by *solving* the above Poisson problem. The obvious part is the final goal: we want to show *existence, uniqueness and stability* of the solution; then, based on these results, we want to *compute* the solution by Numerical Analysis approximation methods.

Less obvious is the *meaning of solution.* In fact, in principle, every problem may be formulated in several ways and a different notion of solution is associated with each way. What is important in the applications is to select the "most efficient notion" for the problem under examination, where "efficiency" may stand for the best compromise between *simplicity of both formulation and theoretical treatment, sufficient flexibility and generality, adaptability to numerical methods.*

Here is a (non exhaustive!) list of various notions of solution for the Poisson problem.

Classical solutions are twice continuously differentiable functions; the differential equation and the boundary conditions are satisfied in the usual pointwise sense.

Strong solutions belong to the Sobolev space $H^2(\Omega)$. Thus, they possess derivatives in $L^2(\Omega)$ up to the second order, in the sense of distributions.

The differential equation is satisfied in the pointwise sense, a.e. with respect to the Lebesgue measure in Ω, while the boundary condition is satisfied in the sense of traces.

Distributional solutions belong to $L^1_{loc}(\Omega)$ and the equation holds in the sense of distributions, i.e.:

$$\int_\Omega \{-u\Delta\varphi + a_0(\mathbf{x})u\varphi\}\, d\mathbf{x} = \int_\Omega f\varphi d\mathbf{x}, \quad \forall\varphi \in \mathcal{D}(\Omega).$$

The boundary condition is satisfied in a very weak sense.

Weak or variational solutions belong to the Sobolev space $H^1(\Omega)$. The boundary value problem is recast within the framework of the abstract variational theory developed in Section 7.4. Often the new formulation represents a version of the *principle of virtual work.*

Clearly, all these notions of solution must be connected by a *coherence principle*, which may be stated as follows: if all the data (domain, boundary data, forcing terms) and the solution are C^∞, *all the above notions must be equivalent.* Thus, the *non-classical* notions constitute a generalization of the classical one.

An important task, with consequences in the error control in numerical methods, is to establish the optimal degree of regularity of a non-classical solution.

More precisely, let u be a non-classical solution of the Poisson problem. The question is: *how much does the regularity of the data a_0, f and of the domain Ω affect the regularity of the solution?*

An exhaustive answer requires rather complicated tools. In the sequel we shall indicate only the most relevant results.

The theory for classical and strong solutions is well established and can be found in specialized books (see the references). From the numerical point of view, the *method of finite differences* best fits the differential structure of the problem and aims at approximating classical solutions.

The distributional theory is well developed, is quite general, but is not the most appropriate framework for solving boundary value problems.

Indeed, the sense in which the boundary values are attained is one of the most delicate points, when one is willing to widen the notion of solution.

For our purposes, the most convenient notion of solution is the last one: it leads to a quite flexible formulation with a sufficiently high degree of generality and a basic theory solely relying on the Lax-Milgram Theorem (Section 7.4). Moreover, the analogy (and often the coincidence) with the principle of virtual work indicates a direct connection with the physical interpretation.

Finally, the variational formulation is the most natural one to implement the *Galerkin method* (*finite elements, spectral elements, etc...*), widely used in the numerical approximation of the solutions of boundary value problems.

To present the main ideas behind the variational formulation, we start from one-dimensional problems with an equation slightly more general than Poisson's equation.

8.3 Diffusion, Drift and Reaction ($n = 1$)

8.3.1 The problem

We shall derive the variational formulation of the following problem:

$$\begin{cases} \underbrace{-(p(x)u')'}_{diffusion} + \underbrace{q(x)\,u'}_{transport} + \underbrace{r(x)u}_{reaction} = f(x)\,, & \text{in the interval } (a,b) \\ \text{boundary conditions} & \text{at } x = a \text{ and } x = b. \end{cases} \tag{8.4}$$

We may interpret (8.4) as a stationary problem of *diffusion, drift and reaction.*

The main steps for the weak formulation are the following:

a) Select a space of *smooth test functions, adapted to the boundary conditions.*

b) Multiply the differential equation by a *test* function and integrate over (a, b).

c) Carry one of the derivatives in the divergence term onto the test function via an integration by parts, using the boundary conditions and obtaining an *integral equation.*

d) Interpret the integral equation as an *abstract variational problem* (Section 7.4) in a suitable Hilbert space. In general, this is a Sobolev space, given by the topological closure of the space of test functions.

8.3.2 Dirichlet conditions

We start by analyzing *homogeneous Dirichlet conditions*:

$$\begin{cases} -(p(x)u')' + q(x)\,u' + r(x)u = f(x)\,, & \text{in } (a,b) \\ u(a) = u(b) = 0. \end{cases} \tag{8.5}$$

Assume first that $p \in C^1([a,b])$, with $p > 0$, and $q, r, f \in C([a,b])$.

Let $u \in C^2(a,b) \cap C([a,b])$ be a classical solution of (8.5). We select $C_0^1(a,b)$ as the space of test functions. These test functions have a continuous derivative and compact support in (a, b). In particular, *they vanish at the end points.*

Now we multiply the equation by an arbitrary $v \in C_0^1(a,b)$ and integrate over (a,b). We find

$$-\int_a^b (pu')'vdx + \int_a^b [qu' + ru]vdx = \int_a^b fvdx. \tag{8.6}$$

Integrating by parts the first term and using $v(a) = v(b) = 0$, we get

$$-\int_a^b (pu')'vdx = \int_a^b pu'v'dx - [pu'v]_a^b = \int_a^b pu'v'dx.$$

From (8.6) we derive the integral equation

$$\int_a^b [pu'v' + qu'v + ruv]dx = \int_a^b fvdx, \qquad \forall v \in C_0^1(a,b). \tag{8.7}$$

Thus, (8.5) implies (8.7).

On the other hand, assume that (8.7) is true. Integrating by parts in the reverse order, we recover (8.6), which can be written in the form

$$\int_a^b \{-(pu')' + q(x)u' + r(x)u - f(x)\} vdx = 0 \qquad \forall v \in C_0^1(a,b).$$

The arbitrariness of v entails[1]

$$-(p(x)u')' + q(x)u' + r(x)u - f(x) = 0 \quad \text{in } (a,b)$$

i.e. the original differential equation.

Thus, **for classical solutions, the two formulations** (8.5) e (8.7) **are equivalent**. Observe that equation (8.7):

- involves *only one derivative* of u;
- makes perfect sense even if p, q, r and f are merely locally integrable;
- transforms (8.5) into an integral equation, valid on an infinite-dimensional space of test functions.

These features lead to the following functional setting:

a) we enlarge the class of test functions to $H_0^1(a,b)$, which is the closure of $C_0^1(a,b)$ in H^1-norm;

b) we look for a solution belonging to $H_0^1(a,b)$, in which the homogeneous Dirichlet conditions are already included.

Thus, the **weak** or **variational** formulation of problem (8.5) is:

Determine $u \in H_0^1(a,b)$ such that

$$\int_a^b \{pu'v' + qu'v + ruv\} dx = \int_a^b fvdx, \qquad \forall v \in H_0^1(a,b). \tag{8.8}$$

[1] If $g \in C([a,b])$ and $\int_a^b gvdx = 0$ for every $v \in C_0^1(a,b)$, then $g \equiv 0$ (exercise).

If we introduce the bilinear form

$$B(u,v) = \int_a^b \{pu'v' + qu'v + ruv\}\, dx$$

and the linear functional

$$Lv = \int_a^b fv\ dx,$$

equation (8.8) can be recast as

$$B(u,v) = Lv, \quad \forall v \in H_0^1(a,b).$$

Then existence, uniqueness and stability follow from the Lax-Milgram Theorem 7.10, under rather natural hypotheses on $p, q, r,\ f$. Recall that, by Poincaré's inequality (7.55) we have

$$\|u\|_0 \le C_P \|u'\|_0,$$

so that we may choose in $H_0^1(a,b)$ the norm

$$\|u\|_1 = \|u'\|_0$$

equivalent to $\|u\|_{1,2} = \|u\|_0 + \|u'\|_0$.

Proposition 8.1. *Assume that $p, q, q', r \in L^\infty(a,b)$ and $f \in L^2(a,b)$. If*

$$p(x) \ge \alpha > 0 \quad \text{and} \quad -\frac{1}{2}q'(x) + r(x) \ge 0 \qquad \text{a.e. in } (a,b), \tag{8.9}$$

then (8.8) has a unique solution $u \in H_0^1(a,b)$. Moreover

$$\|u'\|_0 \le \frac{C_P}{\alpha} \|f\|_0. \tag{8.10}$$

Proof. Let us check that the hypotheses of the Lax-Milgram Theorem hold, with $V = H_0^1(a,b)$.

Continuity of the bilinear form B. We have:

$$|B(u,v)| \le \int_a^b \{\|p\|_{L^\infty} |u'v'| + \|q\|_{L^\infty} |u'v| + \|r\|_{L^\infty} |uv|\}\, dx.$$

Using the Schwarz and Poincaré inequalities, we obtain

$$\begin{aligned} |B(u,v)| &\le \|p\|_{L^\infty} \|u'\|_0 \|v'\|_0 + \|q\|_{L^\infty} \|u'\|_0 \|v\|_0 + \|r\|_{L^\infty} \|u\|_0 \|v\|_0 \\ &\le \left(\|p\|_{L^\infty} + C_P \|q\|_{L^\infty} + C_P^2 \|r\|_{L^\infty}\right) \|u'\|_0 \|v'\|_0 \end{aligned}$$

so that B is continuous in V.

Coerciveness of B. We may write:

$$\begin{aligned} B(u,u) &= \int_a^b \left\{p(u')^2 + qu'u \ + ru^2\right\} dx \\ &\geq \alpha \|u'\|_0^2 + \frac{1}{2}\int_a^b q\left(u^2\right)' dx + \int_a^b ru^2 dx \\ \text{(integrating by parts)} &= \alpha \|u'\|_0^2 + \int_a^b \left\{-\frac{1}{2}q' + r\right\} u^2 dx \\ \text{(from (8.9))} &\geq \alpha \|u'\|_0^2 \end{aligned}$$

and therefore B is $V-$coercive.

Continuity of L in V. The Schwarz and Poincaré inequalities yield

$$|Lv| = \left|\int_a^b fv \ dx\right| \leq \|f\|_0 \|v\|_0 \leq C_P \|f\|_0 \|v'\|_0 .$$

so that $\|L\|_{V^*} \leq C_P \|f\|_0$.

Then, the Lax-Milgram Theorem gives existence, uniqueness and the stability estimate (8.10). □

Remark 8.1. If $q = 0$, the bilinear form B is symmetric. From Theorem 7.11, in this case the weak solution minimizes in $H_0^1(a,b)$ the "energy functional"

$$J(u) = \int_a^b \left\{p(u')^2 + ru^2 - 2fu\right\} dx.$$

Then, equation (8.8) coincides with the Euler equation of J:

$$J'(u)\, v = 0, \qquad \forall v \in H_0^1(a,b).$$

Remark 8.2. In the case of nonhomogeneous Dirichlet conditions, e.g. $u(a) = A$, $u(b) = B$, set $w = u - y$, where $y = y(x)$ is the straight line through the points (a, A), (b, B), given by

$$y(x) = A + (x-a)\frac{B-A}{b-a}.$$

Then, the variational problem for w is

$$\int_a^b [pw'v' + qw'v + rwv]dx = \int_a^b (Fv + Gv')\ dx \qquad \forall v \in H_0^1(a,b) \tag{8.11}$$

with

$$F(x) = f(x) + \frac{B-A}{b-a}q(x) - r(x)\left(A + (x-a)\frac{B-A}{b-a}\right)$$

and

$$G(x) = \frac{B-A}{b-a}p(x).$$

8.3.3 Neumann conditions

We now derive the weak formulation of the Neumann problem

$$\begin{cases} -(p(x)u')' + q(x)u' + r(x)u = f(x), & \text{in } (a,b) \\ -p(a)u'(a) = A, \quad p(b)u'(b) = B. \end{cases} \tag{8.12}$$

The boundary conditions prescribe the outward flux at the end points. This way of writing the Neumann conditions, with the presence of the factor p in front of the derivative, is naturally associated with the divergence structure of the diffusion term.

Again, assume first that $p \in C^1([a,b])$, with $p > 0$, and $q, r, f \in C^0([a,b])$. A classical solution u has a continuous derivative up to the end points so that $u \in C^2(a,b) \cap C^1([a,b])$.

As space of test functions, we choose $C^1([a,b])$. Multiplying the equation by an arbitrary $v \in C^1([a,b])$ and integrating over (a,b), we find again

$$-\int_a^b (pu')'v dx + \int_a^b [qu' + ru]v dx = \int_a^b f v dx. \tag{8.13}$$

Integrating by parts the first term and using the Neumann conditions, we get

$$-\int_a^b (pu')'v dx = \int_a^b pu'v' dx - [pu'v]_a^b = \int_a^b pu'v' dx - v(b)B - v(a)A.$$

Then (8.13) becomes

$$\int_a^b [pu'v' + qu'v + ruv] dx - v(b)B - v(a)A = \int_a^b f v dx, \tag{8.14}$$

for every $v \in C^1([a,b])$.

Thus, (8.12) implies (8.14). If the choice of the test functions is correct, we should be able to recover the classical formulation from (8.14).

Indeed, let us start recovering the differential equation. Since

$$C_0^1(a,b) \subset C^1([a,b]),$$

(8.14) clearly holds for every $v \in C_0^1(a,b)$. Then, (8.14) reduces to (8.7) and we deduce, as before,

$$-(pu')' + qu' + ru - f = 0, \qquad \text{in } (a,b). \tag{8.15}$$

Let us now use the test functions which *do not vanish at the end points.* Integrating by parts the first term in (8.14) we have:

$$\int_a^b pu'v' dx = -\int_a^b (pu')'v \, dx + p(b)v(b)u'(b) - p(a)v(a)u'(a).$$

Inserting this expression into (8.14) and taking into account (8.15) we find:

$$v(b)\left[p(b)u'(b)-B\right]-v(a)\left[p(a)u'(a)+A\right]=0.$$

The arbitrariness of the values $v(b)$ and $v(a)$ forces

$$p(b)u'(b)=B, \qquad -p(a)u'(a)=A,$$

recovering the Neumann conditions as well.

Thus, **for classical solutions, the two formulations** (8.12) and (8.14) **are equivalent**.

Enlarging the class of test functions to $H^1(a,b)$, which is the closure of $C^1([a,b])$ in H^1-norm, we may state the **weak** or **variational** formulation of problem (8.12) as follows:

Determine $u \in H^1(a,b)$ *such that,* $\forall v \in H^1(a,b)$,

$$\int_a^b \{pu'v'+qu'v+ruv\}\,dx=\int_a^b fv\;dx+v(b)B+v(b)A. \tag{8.16}$$

We point out that the Neumann conditions are encoded in equation (8.16), rather than forced by the choice of the test functions, as in the Dirichlet problem.

Introducing the bilinear form

$$B(u,v)=\int_a^b \{pu'v'+qu'v+ruv\}\,dx$$

and the linear functional

$$Lv=\int_a^b fv\;dx+v(b)B+v(a)A,$$

equation (8.16) can be recast in the abstract form

$$B(u,v)=Lv, \quad \forall v \in H^1(a,b).$$

Again, existence, uniqueness and stability of a weak solution follow from the Lax-Milgram Theorem, under rather natural hypotheses on p, q, r, f.

Recall that if $v \in H^1(a,b)$, the Proposition 7.8 yields to the estimate

$$v(x) \le C^* \|v\|_{1,2} \tag{8.17}$$

for every $x \in [a,b]$, with $C^*=\sqrt{2}\max\left\{(b-a)^{-1/2},(b-a)^{1/2}\right\}$.

Proposition 8.2. *Assume that:*

i) $p, q, r \in L^\infty(a, b)$ *and* $f \in L^2(a, b)$*;*

ii) $p(x) \geq \alpha_0 > 0$, $r(x) \geq c_0 > 0$ *a.e. in* (a, b) *and*

$$K_0 \equiv \min\{\alpha_0, c_0\} - \frac{1}{2}\|q\|_{L^\infty} > 0.$$

Then, (8.16) *has a unique solution* $u \in H^1(a, b)$*. Furthermore*

$$\|u\|_{1,2} \leq K_0^{-1}\{\|f\|_0 + C^*(|A| + |B|)\}. \tag{8.18}$$

Proof. Let us check that the hypotheses of the Lax-Milgram Theorem hold, with $V = H^1(a, b)$.

Continuity of the bilinear form B. We have:

$$|B(u, v)| \leq \int_a^b \{\|p\|_{L^\infty}|u'v'| + \|q\|_{L^\infty}|u'v| + \|r\|_{L^\infty}|uv|\}\, dx.$$

Using Schwarz's inequality, we easily get

$$|B(u, v)| \leq (\|p\|_{L^\infty} + \|q\|_{L^\infty} + \|r\|_{L^\infty})\|u\|_{1,2}\|v\|_{1,2}$$

so that B is continuous in V.

Coerciveness of B. We have

$$B(u, u) = \int_a^b \left\{p(u')^2 + qu'u + ru^2\right\} dx.$$

The Schwarz inequality gives

$$\left|\int_a^b qu'u\, dx\right| \leq \|q\|_{L^\infty}\|u'\|_0\|u\|_0 \leq \frac{1}{2}\|q\|_{L^\infty}\left\{\|u'\|_0^2 + \|u\|_0^2\right\}.$$

Then, by *ii*),

$$B(u, u) \geq (\alpha_0 - \frac{1}{2}\|q\|_{L^\infty})\|u'\|_0^2 + (c_0 - \frac{1}{2}\|q\|_{L^\infty})\|u\|_0^2 \geq K_0\|u\|_{1,2}^2$$

so that B is $V-$coercive.

Continuity of L *in* V. Schwarz's inequality and (8.17) yield

$$\begin{aligned}|Lv| &\leq \|f\|_0\|v\|_0 + |v(b)B + v(a)A| \leq \\ &\leq \{\|f\|_0 + C^*(|A| + |B|)\}\|v\|_{1,2}\end{aligned}$$

whence $\|L\|_{V^*} \leq \|f\|_0 + C^*(|A| + |B|)$.

Then, the Lax-Milgram Theorem gives existence, uniqueness and the stability estimate (8.18). □

Remark 8.3. In general, without the condition like $c(x) \geq c_0 > 0$ we cannot expect existence and uniqueness of a solution. In fact, suppose, $p = 1$, $q = r = 0$. The problem reduces to

$$\begin{cases} u'' = f & \text{in } (a,b) \\ -u'(a) = A, \quad u'(b) = B. \end{cases} \tag{8.19}$$

Hypothesis *ii*) is not satisfied (since $r = 0$). If u is a solution of the problem and $k \in \mathbb{R}$, also $u + k$ is a solution of the same problem. We cannot expect uniqueness. Not even we may prescribe f, A, B arbitrarily, if we want that a solution exists. In fact, integrating the equation $u'' = f$ over (a,b), we deduce that the Neumann data and f must satisfy the *compatibility condition*

$$B + A = \int_a^b f(x)\,dx. \tag{8.20}$$

If (8.20) does not hold, *the problem has no solution.* Thus, to have existence and uniqueness we must require that:

a) (8.20) holds;
b) select a solution (e.g.) with zero mean value in (a,b).

Condition (8.20) is indeed quite natural. Problem (8.19), with $A = B = 0$, describes an elastic chord whose end points are free to slide along a vertical guide; in this case (8.20) becomes

$$\int_a^b f(x)\,dx = 0$$

which expresses the obvious fact that in equilibrium conditions the total distributed charge on the chord must vanish.

Remark 8.4. Notice that the Neumann condition is incorporated into the variational formulation of the problem and for this reason they are considered as *natural conditions.* On the opposite, a Dirichlet condition is prescribed at the beginning so that it constitutes a *forced condition.*

8.3.4 Robin and mixed conditions

Suppose that the boundary conditions in problem (8.12) are:

$$-p(a)\,u'(a) = A, \qquad p(b)\,u'(b) + hu(b) = B \quad (h > 0, \text{ constant})$$

where, for simplicity, the Robin condition is imposed at $x = b$ only. With small adjustments, we may repeat the same computations made for the Neumann conditions. The **weak formulation** is:

Determine $u \in H^1(a,b)$ *such that,* $\forall v \in H^1(a,b)$,

$$\int_a^b \{pu'v' + qu'v + ruv\}\,dx + hu(b)\,v(b) = \int_a^b fv\,dx + v(b)\,B + v(a)\,A. \tag{8.21}$$

Introducing the bilinear form

$$\tilde{B}(u,v) = \int_a^b \{pu'v' + qu'v + ruv\}\, dx + hu(b)\, v(b)$$

we may write our problem in the abstract form

$$\tilde{B}(u,v) = Lv \qquad \forall v \in H^1(a,b).$$

We have:

Proposition 8.3. *Assume that i) and ii) of Proposition 8.2 hold and that $h > 0$. Then, (8.21) has a unique solution $u \in H^1(a,b)$. Furthermore*

$$\|u\|_{1,2} \le K_0^{-1} \{\|f\|_0 + C^*(|A| + |B|)\}.$$

Proof. Let $V = H^1(a,b)$. Since

$$\tilde{B}(u,u) = B(u,u) + hu^2(b) \ge K_0 \|u\|_{1,2}^2$$

and

$$\begin{aligned}\left|\tilde{B}(u,v)\right| &\le |B(u,v)| + h\,|u(b)\, v(b)| \\ &\le \left(\|p\|_{L^\infty} + \|q\|_{L^\infty} + \|r\|_{L^\infty} + h(C^*)^2\right) \|u\|_{1,2} \|v\|_{1,2},\end{aligned}$$

$\tilde{B}$ is continuous and $V-$coercive. The conclusion follows easily. □

Mixed conditions. The weak formulation of mixed problems does not present particular difficulties. Suppose, for instance, we assign at the end points the conditions

$$u(a) = 0, \qquad p(b)\, u'(b) = B.$$

Thus, we have a mixed Dirichlet-Neumann problem. The only relevant observation is the choice of the functional setting. Since $u(a) = 0$, we have to choose $V = H^1_{0,a}$, the space of functions $v \in H^1(a,b)$, vanishing at $x = a$. The Poincaré inequality holds in $H^1_{0,a}$, so that we may choose $\|u'\|_0$ as the norm in $H^1_{0,a}$. Moreover, the following inequality

$$v(x) \le C^{**} \|v'\|_0 \tag{8.22}$$

holds[2] for every $x \in [a,b]$, with $C^{**} = (b-a)^{1/2}$.

The **weak formulation** is: *find $u \in H^1_{0,a}$ such that*

$$\int_a^b \{pu'v' + qu'v + ruv\}\, dx = \int_a^b fv dx + v(b)\, B, \qquad \forall v \in H^1_{0,a}. \tag{8.23}$$

[2] Since $v(a) = 0$, we have $v(x) = \int_a^x v'$ so that $|v(x)| \le \sqrt{b-a}\, \|v'\|_0$.